

20260406EB_MN (IF)

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2026年4月6日 星期一

20mM split iPSCs into a full 6-well plate on the Friday before differentiation

differentiate iPSCs in 5-6mL T2 in untreated T25 flask

keep some iPSCs

2026年4月8日 星期三

T2 -> T3

2026年4月10日 星期五

T3 -> T4 (~14 mL for over-weekend)

2026年4月13日 星期一

T4 -> T5

2026年4月15日 星期三

T5 -> T6

2026年4月17日 星期五

T6 -> T6 (~14mL for over-weekend)

Coat PLO

2026年4月19日 星期日

Coat Laminin

2026年4月20日 星期一

T6 -> T7

Dissociation

1. Collect EBs into 50mL tubes (each plate). Wash the plate with 5mL PBS and added them into the tubes. Let EBs settle with gravity.
2. Remove sup. and add 3mL 0.25% trypsin for each (prepare 16mL 0.25% trypsin with 440U DNaseI)
3. Pipette ~8-10 with P1000 and then incubate the tubes in 37C.
4. pipette*10 at 5min, 10min, and 15min. (for iPSC1000, 1066, and 1108, I stop reaction after I pipette all of them at around 17min. I might overdigest them. I should process it right after or just separately)
5. add 6mL FBS into 50mL tube -> pipette with 5mL pipet ~ 10 times
6. add 6mL CTWM -> pipette ~ 10 times
7. filter cells through 40um cell strainer into new tubes.

8. centrifuge at 200xg, 4min

cell info					^
	A	B	C	D	
1	cell	passage	well #	note	
2	WT11	P10	5	low confluency	
3	JIPSC1000	P3+6	3		
4	JIPSC1066	P6+6	3		
5	JIPSC1108	P6+6	3		

Table1			^
	A	B	
1	run1	volume (mL)	
2	trypsin	3	
3	HI-FBS/CTWM	6	

Table3											
	A	B	C	D	E	F	G	H	I	J	K
1		conc. (M/mL)	survival	total volume	tota cell count	plate#	well plate	cell count/well (K)	number of wells	volume	
2	1000	1.62	97	2	3.24	A	24	300	8	1481	
3	1066	1.95	98	2	3.9	A	24	300	8	1231	I thought the calculation might not be correct, so I used the concentration of 1108
4	1108	1.72	98	2	3.44	A	24	300	8	1395	
5	WT11	1.95	99	2	3.9	B	24	279	14		

A							^
	1	2	3	4	5	6	
A	1000		1066		1108		
B							
C							
D							

B							^
	1	2	3	4	5	6	
A				WT11			
B	previous cells						
C							
D							

2026年4月21日 星期二

1000 and 1108 again attached very badly. 1066 is fine. It the same as 20260401MN
I may need to add 2-5 times of cells for these two strains.

2026年4月22日 星期三

1/2 T7-> 1/2 T7

2026年4月23日 星期四

T7 -> T8

2026年4月24日 星期五

1/2 T8 -> 1/2 T8

2026年4月26日 星期日

1/2 T8 -> 1/2 T8

2026年4月28日 星期二

1/2 T8 -> 1/2 T8

2026年4月29日 星期三

IF (treat and fix)

treat cells

condition ^

	A	B	C	D	E	F	G
1	condition#	gluc conc (mM)	media		condition#	MMS	media
2	1	20	no pyruvate or lactate		A	MMS 0mM	with 1mM pyruvate
3	2	7.5	no pyruvate or lactate		B	MMS 1mM	with 1mM pyruvate
4	3	2.5	no pyruvate or lactate		C	MMS 1.5mM	with 1mM pyruvate
5	4	0.5	no pyruvate or lactate		D	MMS 2mM	with 1mM pyruvate
6	5	0.15	no pyruvate or lactate		E	MMS 2.5mM	with 1mM pyruvate
7	6	0.05	no pyruvate or lactate		F	MMS 3mM	with 1mM pyruvate
8	7	0.015	no pyruvate or lactate				
9	8	0	no pyruvate or lactate				

B_layout ^

	1	2	3	4	5	6
A	previous cells			1	5	
B				2	6	
C	A	B	C	3	7	
D	D	E	F	4	8	

drug calculation													
	A	B	C	D	E	F	G	H	I	J	K	L	M
1			stock (mM)	1st dilution factor	total volume for 1st dil.	stock volume	media volume for 1st dil.	conc. after 1st dilution	working (mM)	dilution factor from 1st dil.	total volume	volume of 1st dil.	volume of media
2	MM S	A							0		1000	0	1000
3		B	11800	100	100	1	99	118	1	118	1000	8.5	991.5
4		C	11800	100	100	1	99	118	1.5	78.6666666667	1000	12.7	987.3
5		D	11800	100	100	1	99	118	2	59	1000	16.9	983.1
6		E	11800	100	100	1	99	118	2.5	47.2	1000	21.2	978.8
7		F	11800	100	100	1	99	118	3	39.3333333333	1000	25.4	974.6

prepare media with different conc. of glucose							
	A	B	C	D	E	F	G
1		diluted from (mM)	targetconc. (mM)	dilution factor	final volume (uL)	added volume (uL)	media volume
2	1	1100	20	55	7000	127.3	6872.7
3	2	20	7.5	2.6666666667	4000	1500.0	2500.0
4	3	20	2.5	8	11500	1437.5	10062.5
5	4	2.5	0.5	5	4000	800.0	3200.0
6	5	2.5	0.15	16.6666666667	4000	240.0	3760.0
7	6	2.5	0.05	50	4000	80.0	3920.0
8	7	2.5	0.015	166.666666667	4000	24.0	3976.0
9	8		0		4000	2937.5	1144.0
10				total media volume	43	mL	

2026年5月20日 星期三

Treat HEK cells for virus test (confluent) with 0.5mM or 0.1mM arsenite for 1h as positive controls

Well1						
	1	2	3	4	5	6
A				500uM	100uM	
B				500uM	100uM	
C						
D						

1. Fix HEK cells with 4% PFA
2. wash once with PBS
3. The following steps with MNs
4. 0.2%Triton for 10min at RT (I accidentally treat ~ 10 more minutes for HEK cells)
5. Blocking with 10% FBS in RT for 1h (HEK cells block for ~30min)
6. Primary Ab o/n

Table7													
	A	B	C	D	E	F	G	H	I	J	K	L	M
1	condition	Ab#	antibody	species	dilution	total volume	Ab volume	secondary Ab	#	fluoro	dilution	total volume	Ab volume
2	A	63	H2A.X (Ser139) clone JBW301	Mouse mono	2000	2500	1.25	488	A	488	1000	2700	2.7
3	A	105	G3BP2	Rb	500	2500	5	647	Q	647	1000	2700	2.7
4	backup option	102	TIA1	Rb									

2026年5月21日 星期四

1. PBS wash*3
2. 2nd antibody
3. PBS wash*3
4. check fluorescence
5. DAPI staining
6. PBS wash

2026年5月25日 星期一

Imaging

Notes:

1. HEK cells positive control: B4, 500uM 1h. 100uM 1h didn't form clear stress granules for most cells (see a few cells have a few stress granules)
2. I did not see stress granule formation in MNs
3. There were some "granules" between non-MNs, but I can't tell if those are in the background or in the non-MN cells

NCT_O.C. , 60X, 74X704, 16-bit, old scope

	A	B	C	D	E
1	channel	bit	power (%)	exposure time	note
2	BF	16		40ms	
3	405	16	10	100 ms	
4	488	16	40	300 ms	
5	640	16	10	100ms	

fileID_treatment_IF

	A	B	C	D	E	F
1	image	treatment	treat_time	488	640	
2	C1_488-H2AX_640-G3BP2_405-DAPI	0mM MMS	1h	H2AX	G3BP2	
3	C2_488-H2AX_640-G3BP2_405-DAPI	1mM MMS	1h	H2AX	G3BP2	
4	C3_488-H2AX_640-G3BP2_405-DAPI	1.5mM MMS	1h	H2AX	G3BP2	
5	D1_488-H2AX_640-G3BP2_405-DAPI	2mM MMS	1h	H2AX	G3BP2	
6	D2_488-H2AX_640-G3BP2_405-DAPI	2.5mM MMS	1h	H2AX	G3BP2	XY1 have non-MNs
7	D3_488-H2AX_640-G3BP2_405-DAPI	3mM MMS	1h	H2AX	G3BP2	
8	A4_488-H2AX_640-G3BP2_405-DAPI	20mM gluc	1h	H2AX	G3BP2	
9	A5_488-H2AX_640-G3BP2_405-DAPI	0.15mM gluc	1h	H2AX	G3BP2	XY4 have non-MNs
10	D5_488-H2AX_640-G3BP2_405-DAPI	0mM gluc	1h	H2AX	G3BP2	XY9 have non-MNs